Introduction

IN SEPTEMBER 1962, during the global tumult of the space race, the Cuban missile crisis, and the recently upgraded polio vaccine, there was a less reported—but perhaps equally critical—milestone in human history: It was in the fall of ’62 that we predicted the future.

Cast onto the newly colorful screens of American televisions was the debut of *The Jetsons*, a cartoon about a family living one hundred years in the future. In the guise of a sitcom, the show was, in fact, a prediction of how future humans would live, of what technologies would fill their pockets and furnish their homes.

The Jetsons correctly predicted video calls, flat-screen TVs, cell phones, 3D printing, and smartwatches; all technologies that were unbelievable in 1962 and yet were ubiquitous by 2022. However, there is one technology that we have entirely failed to create, one futurist feat that has not yet come to fruition: the autonomous robot named Rosey.

Rosey was a caretaker for the Jetson family, watching after the children and tending to the home. When Elroy—then six years old—was struggling in school, it was Rosey who helped him with his homework. When their fifteen-year-old daughter, Judy, needed help learning how to drive, it was Rosey who gave her lessons. Rosey cooked meals, set the table, and did the dishes. Rosey was loyal, sensitive, and quick with a joke. She identified brewing family tiffs and misunderstandings, intervening to help individuals see one another’s perspective. At one time, she was moved to tears by a poem Elroy wrote for his mother. Rosey herself, in one episode, even fell in love.

In other words, Rosey had the intelligence of a human. Not just the reasoning, common sense, and motor skills needed to perform complex tasks in the physical world, but also the empathizing, perspective taking, and social finesse needed to successfully navigate our social world. In the words of Jane Jetson, Rosey was "just like one of the family."

Although the *The Jetsons* correctly predicted cell phones and smartwatches, we still don't have anything like Rosey. As of this book going to print, even Rosey's most basic behaviors are still out of reach. It is no secret that the first company to build a robot that can simply *load a dishwasher* will immediately have a bestselling product. All attempts to do this have failed. It isn't fundamentally a *mechanical* problem; it's an *intellectual* one—the ability to identify objects in a sink, pick them up appropriately, and load them without breaking anything has proven far more difficult than previously thought.

Of course, even though we do not yet have Rosey, the progress in the field of artificial intelligence (AI) since 1962 has been remarkable. AI can now beat the best humans in the world at numerous games of skill, including chess and Go. AI can recognize tumors in radiology images as well as human radiologists. AI is on the cusp of autonomously driving cars. And as of the last few years, new advancements in large language models are enabling products like ChatGPT, which launched in fall 2022, to compose poetry, translate between languages at will, and even write code. To the chagrin of every high school teacher on planet Earth, ChatGPT can instantly compose a remarkably well written and original essay on almost any topic that an intrepid student might ask of it. ChatGPT can even pass the bar exam, scoring better than 90 percent of lawyers.

Across this long arrow of AI achievements, it has always been hard to tell how close we are to creating human-level intelligence. After the early successes of problem solving algorithms in the 1960s, the AI pioneer Marvin Minsky famously proclaimed that "from three to eight years we will have a machine with the general intelligence of an average human being." It did not happen. After the successes of expert systems in the 1980s, *BusinessWeek* proclaimed "AI: it's here." Progress stalled shortly thereafter. And now with advancements in large language models, many researchers have again proclaimed that the "game is over" because we are "on the verge of achieving human-level AI." Which is it: Are we finally on the cusp of creating human-like artificial intelligence like Rosey, or are large language models like ChatGPT just the most recent achievement in a long journey that will stretch on for decades to come?

Along this journey, as AI keeps getting smarter, it is becoming harder to measure our progress toward this goal. If an AI system outperforms humans on a task, does it mean that the AI system has captured how humans solve the task? Does a calculator—capable of crunching numbers faster than a human—actually understand math? Does ChatGPT—scoring better on the bar exam than most lawyers—actually understand the law? How can we tell the difference, and in what circumstances, if any, does the difference even matter?

In 2021, over a year before the release of ChatGPT—the chatbot that is now rapidly proliferating throughout every nook and cranny of society—I was using its precursor, a large language model called GPT-3. GPT-3 was trained on large quantities of text (large as in *the entire internet*), and then used this corpus to try to pattern match the most likely response to a prompt. When asked, “What are two reasons that a dog might be in a bad mood?” it responded, “Two reasons a dog might be in a bad mood are if it is hungry or if it is hot.” Something about the new architecture of these systems enabled them to answer questions with what at least seemed like a remarkable degree of intelligence. These models were able to generalize facts they had read about (like the Wikipedia pages about dogs and other pages about causes of bad moods) to new questions they had never seen. In 2021, I was exploring possible applications of these new language models—could they be used to provide new support systems for mental health, or more seamless customer service, or more democratized access to medical information?

The more I interacted with GPT-3, the more mesmerized I became by both its successes and mistakes. In some ways it was brilliant, in other ways it was oddly dumb. Ask GPT-3 to write an essay about eighteenth-century potato farming and its relationship to globalization, and you will get a surprisingly coherent essay. Ask it a commonsense question about what someone might see in a basement, and it answers nonsensically.[*] Why could GPT-3 correctly answer some questions and not others? What features of human intelligence does it capture, and which is it missing? And why, as AI development continues to accelerate, are some questions that were hard to answer in one year becoming easy in subsequent years? Indeed, as of this book going to print, the new and upgraded version of GPT-3, called GPT-4, released in early 2023, can correctly answer many questions that beguiled GPT-3. And yet still, as we will see in this book, GPT-4 fails to capture essential features of human intelligence—about something going on in the human brain.

Indeed, the discrepancies between artificial intelligence and human intelligence are nothing short of perplexing. Why is it that AI can crush any human on earth in a game of chess but can't load a dishwasher better than a six-year-old?

We struggle to answer these questions because we don't yet understand the thing we are trying to re-create. All of these questions are, in essence, not questions about AI, but about the nature of human intelligence itself—how it works, why it works the way it does, and as we will soon see, most importantly, how it came to be.

Nature's Hint

When humanity wanted to understand flight, we garnered our first inspiration from birds; when George de Mestral invented Velcro, he got the idea from burdock fruits; when Benjamin Franklin sought to explore electricity, his first sparks of understanding came from lightning. Nature has, throughout the history of human innovation, long been a wondrous guide.

Nature also offers us clues as to how intelligence works—the clearest locus of which is, of course, the human brain. But in this way, AI is unlike these other technological innovations; the brain has proven to be more unwieldy and harder to decipher than either wings or lightning. Scientists have been investigating how the brain works for millennia, and while we have made progress, we do not yet have satisfying answers.

The problem is complexity.

The human brain contains eighty-six billion neurons and over a hundred trillion connections. Each of those connections is so minuscule—less than thirty nanometers wide—that they can barely be seen under even the most powerful microscopes. These connections are bunched together in a tangled mess—within a single cubic millimeter (the width of a single *letter* on a penny), there are over *one billion* connections.

But the sheer number of connections is only one aspect of what makes the brain complex; even if we mapped the wiring of each neuron we would still be far from understanding how the brain works. Unlike the electrical connections in your computer, where wires all communicate using the same signal—electrons—across each of these neural connections, hundreds of different chemicals are passed, each with completely different effects. The simple fact that two neurons connect to each other tells us little about what they are communicating. And worst of all, these connections themselves are in a constant state of change, with some neurons branching out and forming new connections, while others are retracting and removing old ones. Altogether, this makes reverse engineering how the brain works an ungodly task.

Studying the brain is both tantalizing and infuriating. One inch behind your eyes is the most awe-inspiring marvel of the universe. It houses the secrets to the nature of intelligence, to building humanlike artificial intelligence, to why we humans think and behave the way we do. It is right there, reconstructed millions of times per year with every newly born human. We can touch it, hold it, dissect it, we are *literally made of it*, and yet its secrets remain out of reach, hidden in plain sight.

If we want to reverse-engineer how the brain works, if we want to build Rosey, if we want to uncover the hidden nature of human intelligence, perhaps the human brain is not nature's best clue. While the most intuitive place to look to understand the human brain is, naturally, inside the human brain itself, counterintuitively, this may be the *last* place to look. The best place to start may be in dusty fossils deep in the Earth's crust, in microscopic genes tucked away inside cells throughout the animal kingdom, and in the brains of the many *other* animals that populate our planet.

In other words, the answer might not be in the present, but in the hidden remnants of a long time past.

The Missing Museum of Brains

> *I have always been convinced that the only way to get artificial intelligence to work is to do the computation in a way similar to the human brain.*
>
> —GEOFFREY HINTON (PROFESSOR AT UNIVERSITY OF TORONTO, CONSIDERED ONE OF THE "GODFATHERS OF AI")

Humans fly spaceships, split atoms, and edit genes. No other animal has even invented the wheel.

Because of humanity's larger résumé of inventions, you might think that we would have little to learn from the brains of other animals. You might think that the human brain would be entirely unique and nothing like the brains of other animals, that some special brain structure would be the secret to our cleverness. But this is not what we see.

What is most striking when we examine the brains of other animals is how remarkably *similar* their brains are to our own. The difference between our brain and a chimpanzee's brain, besides size, is barely anything. The difference between our brain and a rat's brain is only a handful of brain modifications. The brain of a fish has almost all the same structures as our brain.

These similarities in brains across the animal kingdom mean something important. They are clues. Clues about the nature of intelligence. Clues about ourselves. Clues about our past.

Although today brains are complex, they were not always so. The brain emerged from the unthinking chaotic process of evolution; small random variations in traits were selected for or pruned away depending on whether they supported the further reproduction of the life-form.

In evolution, systems start simple, and complexity emerges only over time.[*] The first brain—the first collection of neurons in the head of an animal—appeared six hundred million years ago in a worm the size of a grain of rice. This worm was the ancestor of all modern brain-endowed animals. Over hundreds of millions of years of evolutionary tinkering, through trillions of small tweaks in wiring, her simple brain was transformed into the diverse portfolio of modern brains. One lineage of this ancient worm's descendants led to the brain in our heads.

If only we could go back in time and examine this first brain to understand how it worked and what tricks it enabled. If only we could then track the complexification forward in the lineage that led to the human brain, observing each physical modification that occurred and the intellectual abilities it afforded. If we could do this, we might be able to grasp the complexity that eventually emerged. Indeed, as the biologist Theodosius Dobzhansky famously said, "Nothing in biology makes sense except in the light of evolution."

Even Darwin fantasized about reconstructing such a story. He ends his *Origin of Species* fantasizing about a future when "psychology will be based on a new foundation, that of the necessary acquirement of each mental power and capacity by gradation." One hundred fifty years after Darwin, this may finally be possible.

Although we have no time machines, we can, in principle, engage in time travel. In just the past decade, evolutionary neuroscientists have made incredible progress in reconstructing the brains of our ancestors. One way they do this is through the fossil record—scientists can use the fossilized skulls of ancient creatures to reverse-engineer the structure of their brains. Another way to reconstruct the brains of our ancestors is by examining the brains of other animals in the animal kingdom.

The reason why brains across the animal kingdom are so similar is that they all derive from common roots in shared ancestors. Every brain in the animal kingdom is a little clue as to what the brains of our ancestors looked like; each brain is not only a machine but a time capsule filled with hidden hints of the trillions of minds that came before. And by examining the intellectual feats these other animals share and those they do not, we can begin to not only reconstruct the brains of our ancestors, but also determine what intellectual abilities these ancient brains afforded them. Together, we can begin to trace acquirement of each mental power by gradation.

It is all, of course, still a work in progress, but the story is becoming tantalizingly clear.

The Myth of Layers

I am hardly the first to propose an evolutionary framework for understanding the human brain. There is a long tradition of such frameworks. The most famous was formulated in the 1960s by the neuroscientist Paul MacLean. MacLean hypothesized that the human brain was made of three layers (hence *triune*), each built on top of another: the *neocortex*, which evolved most recently, on top of the *limbic system*, which evolved earlier, on top of the *reptile brain,* which evolved first.

MacLean argued that the reptile brain was the center of our basic survival instincts, such as aggression and territoriality. The limbic system was supposedly the center of emotions, such as fear, parental attachment, sexual desire, and hunger. And the neocortex was supposedly the center of cognition, gifting us with language, abstraction, planning, and perception. MacLean's framework suggested that reptiles had *only* a reptile brain, mammals like rats and rabbits had a reptile brain *and* a limbic system, and we humans had all three systems. Indeed, to him, these “three evolutionary formations might be imagined as three interconnected biological computers, with each having its own special intelligence, its own subjectivity, its own sense of time and space, and its own memory, motor, and other functions.”

The problem is that MacLean’s Triune Brain Hypothesis has been largely discredited—not because it is inexact (all frameworks are inexact), but because it leads to the wrong conclusions about how the brain evolved and how it works. The implied brain anatomy is wrong; the brains of reptiles are not only made up of the structures MacLean referred to as the “reptile brain”; reptiles also have their own version of a limbic system. The functional divisions proved wrong; *survival instincts*, *emotions*, and *cognition* do not delineate cleanly—they emerge from diverse networks of systems spanning all three of these supposed layers. And the implied evolutionary story turned out to be wrong. You do not have a reptile brain in your head; evolution did not work by simply layering one system on top of another without any modifications to the existing systems.

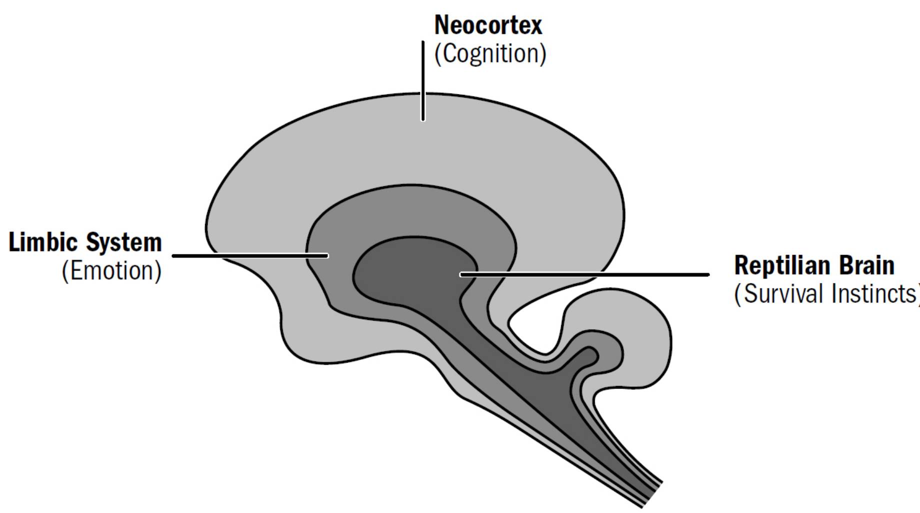

Figure 1: MacLean’s triune brain

Figure by Max Bennett (inspired by similar figures found in MacLean’s work)

But even if MacLean’s triune brain had turned out to be closer to the truth, its biggest problem is that its functional divisions aren’t particularly useful for our purposes. If our goal is to reverse-engineer the human brain to understand the nature of intelligence, MacLean’s three systems are too broad and the functions attributed to them too vague to provide us with even a point at which to start.

We need to ground our understanding of how the brain works and how it evolved in our understanding of how intelligence works—for which we must look to the field of artificial intelligence. The relationship between AI and the brain goes both ways; while the brain can surely teach us much about how to create artificial humanlike intelligence, AI can also teach us about the brain. If we think some part of the brain uses some specific algorithm but that algorithm doesn't work when we implement it in machines, this gives us evidence that the brain might not work this way. Conversely, if we find an algorithm that works well in AI systems, and we find parallels between the properties of these algorithms and properties of animal brains, this gives us some evidence that the brain might indeed work this way.

The physicist Richard Feynman left the following on a blackboard shortly before his death: "What I cannot create, I do not understand." The brain is our guiding inspiration for how to build AI, and AI is our litmus test for how well we understand the brain.

We need a new evolutionary story of the brain, one grounded not only in a modern understanding of how brain anatomy changed over time, but also in a modern understanding of intelligence itself.

The Five Breakthroughs

> *Let's start with Artificial Rat–level intelligence (ARI), then move on to Artificial Cat–level intelligence (ACI), and so on to Artificial Human–level Intelligence (AHI).*
>
> —YANN LECUN, HEAD OF AI AT META

We have a lot of evolutionary history to cover—four billion years. Instead of chronicling each minor adjustment, we will be chronicling the major evolutionary breakthroughs. In fact, as an initial approximation—a first template of this story—the entirety of the human brain's evolution can be reasonably summarized as the culmination of only *five* breakthroughs, starting from the very first brains and going all the way to human brains.

These five breakthroughs are the organizing map to our book, and they make up our itinerary for our adventure back in time. Each breakthrough emerged from new sets of brain modifications and equipped animals with a new portfolio of intellectual abilities. This book is divided into five parts, one for each breakthrough. In each section, I will describe why these abilities evolved, how they worked, and how they still manifest in human brains today.

Each subsequent breakthrough was built on the foundation of those that came before and provided the foundation for those that would follow. Past innovations enabled future innovations. It is through this ordered set of modifications that the evolutionary story of the brain helps us make sense of the complexity that eventually emerged.

But this story cannot be faithfully retold by considering only the biology of our ancestors' brains. These breakthroughs always emerged from periods when our ancestors faced extreme situations or got caught in powerful feedback loops. It was these pressures that led to rapid reconfigurations of brains. We cannot understand the breakthroughs in brain evolution without also understanding the trials and triumphs of our ancestors: the predators they outwitted, the environmental calamities they endured, and the desperate niches they turned to for survival.

And crucially, we will ground these breakthroughs in what is currently known in the field of AI, for many of these breakthroughs in biological intelligence have parallels to what we have learned in artificial intelligence. Some of these breakthroughs represent intellectual tricks we understand well in AI, while other tricks still lay beyond our understanding. And in this way, perhaps the evolutionary story of the brain can shed light on what breakthroughs we may have missed in the development of artificial humanlike intelligence. Perhaps it will reveal some of nature's hidden clues.

Me

I wish I could tell you that I wrote this book because I have spent my whole life pondering the evolution of the brain and trying to build intelligent robots. But I am not a neuroscientist or a roboticist or even a scientist. I wrote this book because I wanted to read this book.

I came to the perplexing discrepancy between human and artificial intelligence by trying to apply AI systems to real-world problems. I spent the bulk of my career at a company I cofounded named Bluecore; we built software and AI systems to help some of the largest brands in the world personalize their marketing. Our software helped predict what consumers would buy before they knew what they wanted. We were merely one tiny part in a sea of countless companies beginning to use the new advances in AI systems. But all these many projects, both big and small, were shaped by the same perplexing questions.

When commercializing AI systems, there is eventually a series of meetings between business teams and machine learning teams. The business teams look for applications of new AI systems that would be *valuable*, while only the machine learning teams understand what applications would be *feasible*. These meetings often reveal our mistaken intuitions about how much we understand about intelligence. Businesspeople probe for applications of AI systems that seem straightforward to them. But frequently, these tasks seem straightforward only because they are straightforward for *our brains.* Machine learning people then patiently explain to the business team why the idea that seems simple is, in fact, astronomically difficult. And these debates go back and forth with every new project. It was from these explorations into how far we could stretch modern AI systems and the surprising places where they fall short that I developed my original curiosity about the brain.

Of course, I am also a human and I, like you, have a human brain. So it was easy for me to become fascinated with the organ that defines so much of the human experience. The brain offers answers not only about the nature of intelligence, but also why we behave the way we do. Why do we frequently make irrational and self-defeating choices? Why does our species have such a long recurring history of both inspiring selflessness and unfathomable cruelty?

My personal project began with merely trying to read books to answer my own questions. This eventually escalated to lengthy email correspondences with neuroscientists who were generous enough to indulge the curiosities of an outsider. This research and these correspondences eventually led me to publish several research papers, which all culminated in the decision to take time off work to turn these brewing ideas into a book.

Throughout this process, the deeper I went, the more I became convinced that there was a worthwhile synthesis to be contributed, one that could provide an accessible introduction to how the brain works, why it works the way it does, and how it overlaps and differs from modern AI systems; one that could bring various ideas across neuroscience and AI together under an umbrella of a single story.

A Brief History of Intelligence is a synthesis of the work of many others. At its heart, it is merely an attempt to put together the pieces that were already there. I have done my best to give due credit throughout the book, always aiming to celebrate those scientists who did the actual research. Anywhere I have failed to do so is unintentional. Admittedly, I couldn't resist sprinkling in a few speculations of my own, but I will aim to be clear when I step into such territory.

It is perhaps fitting that the origin of this book, like the origin of the brain itself, came not from prior planning but from a chaotic process of false starts and wrong turns, from chance, iteration, and lucky circumstance.

A Final Point (About Ladders and Chauvinism)

I have one final point to make before we begin our journey back in time. There is a misinterpretation that will loom dangerously between the lines of this entire story.

This book will draw many comparisons between the abilities of humans and those of other animals alive today, but this is always done by picking specifically those animals that are believed to be most similar to our ancestors. This entire book—the five-breakthroughs framework itself—is solely the story of the *human* lineage, the story of how *our* brains came to be; one could just as easily construct a story of how the octopus or honeybee brain came to be, and it would have its own twists and turns and its own breakthroughs.

Just because our brains wield more intellectual abilities than those of our ancestors does not mean that the modern human brain is strictly intellectually superior to those of other modern animals.

Evolution independently converges on common solutions all the time. The innovation of wings independently evolved in insects, bats, and birds; the common ancestor of these creatures did not have wings. Eyes are also believed to have independently evolved many times. Thus, when I argue that an intellectual ability, such as episodic memory, evolved in early mammals, this does *not* mean that today *only* mammals have episodic memory. Like with wings and eyes, other lineages of life may have independently evolved episodic memory. Indeed, many of the intellectual faculties that we will chronicle in this book are not unique to our lineage, but have independently sprouted along numerous branches of earth's evolutionary tree.

Since the days of Aristotle, scientists and philosophers have constructed what modern biologists refer to as a "scale of nature" (or, since scientists like using Latin terms, *scala naturae*). Aristotle created a hierarchy of all life-forms with humans being superior to other mammals, who were in turn superior to reptiles and fish, who were in turn superior to insects, who were in turn superior to plants.

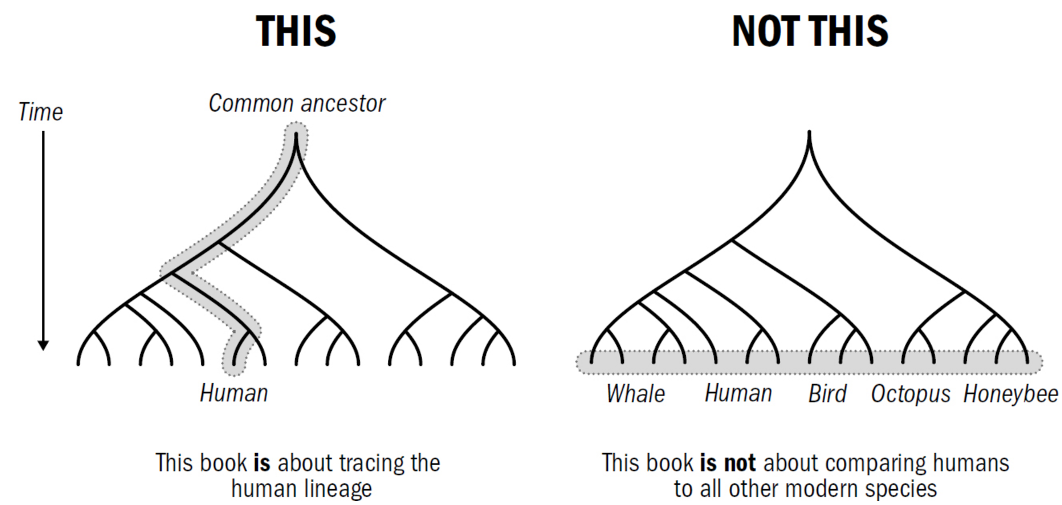

Figure 2

Original art by Rebecca Gelernter

Even after the discovery of evolution, the idea of a scale of nature continues to persist. This idea that there is a hierarchy of species is dead wrong. All species alive today are, well, *alive*; their ancestors survived the last 3.5 billion years of evolution. And thus, in that sense—the only sense that evolution cares about—all life-forms alive today are tied for first place.

Species fall into different survival niches, each of which optimizes for different things. Many niches—in fact, *most* niches—are better served by *smaller* and *simpler* brains (or no brains at all). Big-brained apes are the result of a different survival strategy than that of worms, bacteria, or butterflies. But none are "better." In the eyes of evolution, the hierarchy has only two rungs: on one, there are those that survived, and on the other, those that did not.

Perhaps instead, one wants to define *better* by some specific feature of intelligence. But here still, the ranking will entirely depend on what specific intellectual skill we are measuring. An octopus has an independent brain in each of its tentacles and can blow a human away at multitasking. Pigeons, chipmunks, tuna, and even *iguanas* can process visual information faster than a human. Fish have incredibly accurate real-time processing; have you ever seen how fast a fish whips through a maze of rocks if you try to grab it? A human would surely crash if he or she tried to move so quickly through an obstacle course.

My appeal: As we trace our story, we must avoid thinking that the complexification from past to future suggests that modern humans are strictly superior to modern animals. We must avoid the accidental construction of a *scala naturae*. All animals alive today have been undergoing evolution for the same amount of time.

However, there are, of course, things that make us humans unique, and because *we are human*, it makes sense that we hold a special interest in understanding ourselves, and it makes sense that we strive to make artificial *humanlike* intelligences. So I hope we can engage in a human-centered story without devolving into human chauvinism. There is an equally valid story to be told for any other animal, from honeybees to parrots to octopuses, with which we share our planet. But we will not tell these stories here. This book tells the story of only *one* of these intelligences: it tells the story of us.